

Conditional Probability

$A, B \Rightarrow$ Events

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

given, $P(B) \neq 0$

examples

$D_1, D_2 \Rightarrow$ 2 dice

$\{(1,1), (1,2), (1,3), \dots, (6,6)\}$

Total 36 samples

sample space

$D_1 \rightarrow$ $D_2 \downarrow$	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9

4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

1) What is the probability of D_1 is 5

$$\therefore P(D_1 = 5) = \frac{1}{36}$$

2) What is prop of $D_1 + D_2 \leq 10$

$$\therefore P(D_1 + D_2 \leq 10) = \frac{33}{36} = \frac{11}{12}$$

Conditional Probability examples

$D_1 = 5$ Given $D_1 + D_2 \leq 10$

↑

A

↑

B

$$P(D_1 = 5 \mid D_1 + D_2 \leq 10)$$

From above chart $\Rightarrow 6, 7, 8, 9, 10$

$$P(A|B) = \frac{5}{33}$$

Using formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$A = \{D_1 = 5\}, \quad B = \{D_1 + D_2 \leq 10\}$$

$$P(A \cap B) = \frac{5}{36}, \quad P(B) = \frac{33}{36}$$

$$\therefore P(A|B) = \frac{\frac{5}{\cancel{36}}}{\frac{33}{\cancel{36}}} = \frac{5}{33}$$